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Lipotoxicity and Oocyte Quality in Mammals: Pathogenesis, Consequences, and Reversibility

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Keywords

lipotoxicity, oocyte, embryo, mitochondria, epigenetics, therapy

Abstract

Metabolic stress conditions are often characterized by upregulated lipolysis and subsequently increased serum free fatty acid (FFA) concentrations, leading to the uptake of FFAs by non-adipose tissues and impairment of their function. This phenomenon is known as lipotoxicity. The increased serum FFA concentrations are reflected in the ovarian follicular fluid, which can have harmful effects on oocyte development. Several studies using in vitro and in vivo mammalian models showed that altered oocyte metabolism, increased oxidative stress, and mitochondrial dysfunction are crucial mechanisms underlying this detrimental impact. Ultimately, this can impair offspring health through the persistence of defective mitochondria in the embryo, hampering epigenetic reprogramming and early development. In vitro and in vivo treatments to enhance oocyte mitochondrial function are increasingly being developed. This can help to improve pregnancy rates and safeguard offspring health in metabolically compromised individuals.

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1. INTRODUCTION

The prevalence of metabolic health disorders in humans is increasing rapidly. According to the latest World Health Organization report (1), the number of adults suffering from obesity has more than doubled over the past 30 years. Currently, 43% of adults are overweight [body mass index (BMI) >25 and <30], and 16% are obese (BMI >30). This is often the result of a high-fat diet (HFD), also termed a Western-type diet, consisting of more than 30% fats (2). In women, overweight and obesity are associated with reduced fertility, in which impaired oocyte quality plays an important primary role (3). However, metabolic disorders can also develop irrespective of the diet. For example, diabetes types 1 and 2 can be related to genetic predisposition, low physical activity, aging, and environmental factors (4). The worldwide number of adults with diabetes has increased from 151 million in 2000 to 537 million in 2021 and is expected to further increase to 783 million in 2045 (5). Diabetes is associated with impaired oocyte quality and reduced pregnancy outcomes in women of reproductive age (6). Other mammalian species may also experience demanding metabolic conditions that can negatively impact oocyte quality and fertility, with the prevalence of these conditions increasing in some cases. For example, the average milk yield per cow has more than doubled over the past 50 years in high-yielding dairy cows (7). Concomitantly, the mobilization of fats that are required to produce this milk has also increased, putting the metabolism of the lactating cow under severe pressure (8). Ultimately, this can result in reduced pregnancy rates (9).

These metabolic disorders all have in common that they will lead to lipotoxicity, which involves the deleterious effects of lipid accumulation in non-adipose tissues such as the ovarian follicle and developing oocyte (10). It is generally accepted that this results in reduced oocyte quality, subfertility, and offspring health defects (11). This review first aims to provide a comprehensive overview of the involved mechanisms and their consequences. Second, it aims to offer insights aiding the development of novel strategies for preserving oocyte quality, enhancing fertility, and safeguarding offspring health in metabolically compromised individuals.

2. METABOLIC HEALTH AND THE LIPOTOXIC STATE

2.1. Pathogenesis of Increased Free Fatty Acid Provision in Non-Adipose Tissues

Long-term consumption of a HFD can lead to obesity and is associated with increased lipid concentrations in the bloodstream, also known as hyperlipidemia. More specifically, it can result in the accumulation of chylomicrons containing dietary fats, very low-density lipoproteins (VLDLs) containing hepatic fats, and low-density lipoproteins (LDLs) that are formed through the metabolism of chylomicron remnants and VLDL particles (12). Lipoproteins are delivered to the tissues, where lipoprotein lipase breaks the triglycerides (TGs) in the lipoproteins down into free fatty acids (FFAs). This causes an increase in FFA uptake mainly in adipose tissue (13). Adipocytes can convert FFAs back into TGs to be stored in lipid droplets. However, under these circumstances adipocytes will eventually become hypertrophic. The increased size is associated with defects in the trafficking of glucose transporter GLUT4 to the plasma membrane, leading to impaired insulin-dependent glucose uptake (14). Hypertrophy of adipocytes also results in a cascade of effects, including macrophage infiltration, and subsequent synthesis of proinflammatory cytokines that impair insulin sensitivity. Furthermore, the secretion of adipokines by the adipocytes promotes additional activation of macrophages and the synthesis of proinflammatory cytokines, further aggravating reduced insulin sensitivity in adipose tissue. Because insulin has a strong anti-lipolytic effect, the culmination of these factors ultimately leads to increased lipolysis, resulting in elevated serum FFA concentrations and lipotoxicity (15). Lipotoxicity also commonly

occurs in various other metabolic disorders, like type 1 and 2 diabetes, and metabolic syndrome (16). In these conditions, the reduction in insulin levels or insulin sensitivity leads to ineffective inhibition of hormone-sensitive lipase, increasing lipolysis in adipose tissue (17).

Interestingly, lipotoxicity not only is associated with HFD consumption or an insulin-resistant state but also is relevant in cases of restricted food intake, weight loss, or increased metabolic demands that exceed energy provision (18). For example, in high-producing dairy cows, ingested nutrients are preferentially distributed to the offspring during the final weeks of pregnancy and to milk production during the postpartum (pp) period. The cow cannot compensate for the high energetic demands via food intake. Consequently, high-producing dairy cows develop a state of negative energy balance (NEB) after calving, during which they lose weight. This involves a low insulin state and increased lipolysis from fat reserves to mobilize energy, resulting in elevated serum FFA concentrations (8). In severe cases of NEB, with a strong reduction in body condition score, this is associated with lipotoxicity (9). Similar problems have been observed in sows suffering from lactation weight loss (19). Therefore, the dairy cow and pig are considered relevant model species to investigate lipotoxicity and its effects on, for example, oocyte quality (20, 21). In summary, although the pathophysiology is different between the discussed conditions, they can all give rise to increased, lipotoxic FFA provision in non-adipose tissues.

2.2. Cellular Uptake of Free Fatty Acids

The increased concentration of circulating FFAs is bound to albumin and can be taken up directly by cells in non-adipose tissues, whereas lipids in lipoproteins first need to be broken down by lipoprotein lipase to release FFAs. Three main types of fatty acid transporters are responsible for cellular FFA uptake: cluster of differentiation 36 (CD36), fatty acid binding protein, and fatty acid transport protein (16) (**Figure 1**). In the cytosol, FFAs are converted to acyl-CoA by acyl-CoA synthetase long-chain family member 1. Carnitine palmitoyl transferase-1 and -2 (CPT-1 and -2) transfer acyl-CoA to the mitochondria, where it is metabolized via β-oxidation. This initiates the production of acetyl-CoA, which is then further metabolized in the tricarboxylic acid (TCA) cycle (22). When too much acetyl-CoA is produced to be metabolized in the TCA cycle, this is transported to the cytosol via carnitine acetyltransferase. Also, citrate that is produced in the TCA cycle can be converted to cytosolic acetyl-CoA by ATP-citrate lyase (23). Then, cytosolic acetyl-CoA is converted into malonyl-CoA by acetyl-CoA carboxylase-2. When excess malonyl-CoA is formed, the transport of long-chain fatty acids as acyl-CoA into the mitochondria is inhibited (24). Moreover, in lipotoxic conditions, the capacity of CPT-1 and -2 to transfer acyl-CoA to the mitochondrial (MT) matrix is exceeded. For these reasons, acyl-CoA concentrations increase in the cytosol and are converted into cytosolic TGs and other lipids within lipid droplets. However, when the cytosolic acyl-CoA concentrations become too high, the intrinsic capacity of the cell to esterify the fatty acids is also exceeded, and thus the FFAs remain in the cytosol (16).

2.3. Detrimental Effects on Cell Function Leading to Lipotoxic Effects

In the mitochondria, β-oxidation and the TCA cycle supply $FADH_2$ and NADH to the electron transport chain, where ATP is produced through oxidative phosphorylation (OXPHOS) (25). By-products of OXPHOS are reactive oxygen species (ROS), which play a key role as signaling molecules. Nevertheless, in the lipotoxic state, the metabolization of FFAs increases the rate of OXPHOS, leading to excessive production of ROS. At such high concentrations, ROS are harmful to several cellular structures and functions (26). ROS can induce the oxidation of lipids, for example, in the membrane of the cell or mitochondria, which results in the formation of highly

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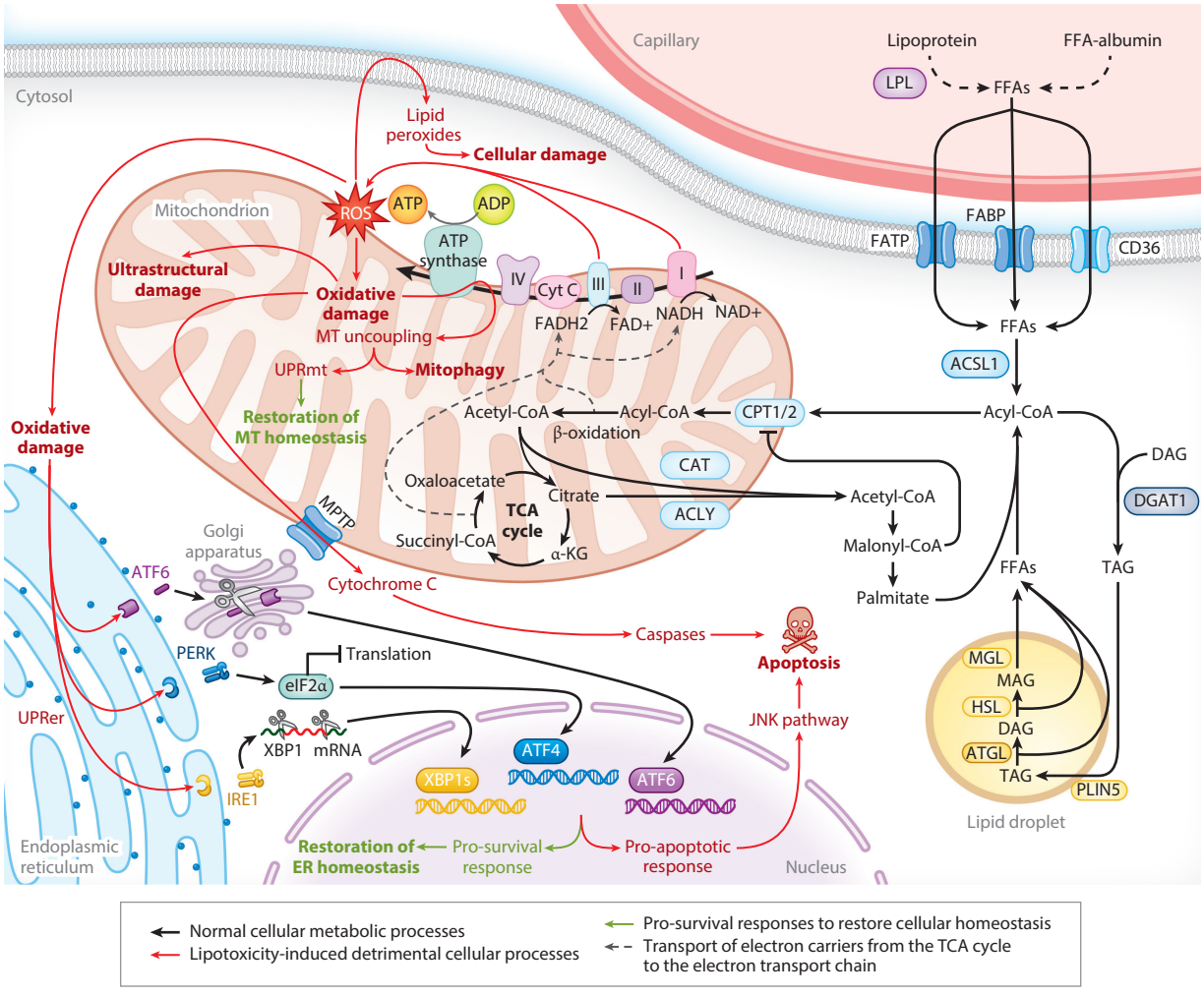

Figure 1

Schematic overview of the cellular impact of lipotoxicity. Increased uptake of FFAs into non-adipose tissue can increase ROS production, leading to several detrimental effects on organelle functions, particularly on the mitochondria and ER. Pro-survival unfolded protein responses in the mitochondria and ER may restore cellular homeostasis under low levels of oxidative stress. Eventually, high levels of stress or prolonged exposure to stress may elicit apoptosis. Abbreviations: α-KG, α-ketoglutarate; ACLY, ATP-citrate lyase; ACSL1, acyl-CoA synthetase long-chain family member 1; ATF4/6, activating transcription factor 4/6; ATGL, adipose triglyceride lipase; CAT, catalase; CPT1/2, carnitine palmitoyl transferase 1/2; DAG, diacylglycerol; DGAT1, O-acyltransferase diacylglycerol; eIF2α, eukaryotic translation initiation factor 2α; ER, endoplasmic reticulum; FABP, fatty acid binding protein; FATP, fatty acid transport protein; FFA, free fatty acid; HSL, hormone-sensitive lipase; IRE1, inositol-requiring enzyme type 1; LPL, lipoprotein lipase; MAG, monoacylglycerol; MGL, monoacylglycerol lipase; MPTP, mitochondrial permeability transition pore; MT, mitochondrial; PERK, protein kinase RNA-like endoplasmic reticulum kinase; PLIN5, perilipin-5; ROS, reactive oxygen species; TAG, triacylglycerol; TCA, tricarboxylic acid; UPRer, endoplasmic reticulum unfolded protein response; UPRmt, mitochondrial unfolded protein response; XBP1, X-box binding protein 1. Figure adapted from "Lipids and Proteins Involved in Lipid Uptake and Metabolism in Cardiac Lipotoxicity." Retrieved from **https://app.biorender.com/biorender-templates**.

reactive lipid peroxides. These can then further disrupt several cellular processes (27). High ROS concentrations can also generate structural alterations to the endoplasmic reticulum (ER), leading to the misfolding of proteins that accumulate and elicit specific ER unfolded protein responses (28). During this response, the binding immunoglobulin protein binds to the unfolded/misfolded

proteins and dissociates from the inositol-requiring enzyme 1, activating transcription factor 6, and protein kinase R-like ER kinase transmembrane proteins, leading to their activation. These factors are translocated to the nucleus to increase the expression of redox regulatory proteins and chaperones. This induces increased protein folding and degrading of misfolded proteins, ultimately restoring ER homeostasis (29). However, in very severe cases homeostasis cannot be restored, resulting in the activation of the c-Jun N-terminal kinase pathway and apoptosis (30).

ROS can also damage the mitochondria itself, although the mitochondria possess several adaptive mechanisms to cope with this oxidative stress (31). When ROS levels are elevated beyond a certain threshold, uncoupling proteins in the inner MT membrane are activated and partially dissipate the proton gradient in the mitochondria. This generates a bypass of electron transport from complex V (or ATP synthase) from the electron transport chain, reducing the driving force of MT ROS production (32). Also, this reduction in the MT membrane potential (MMP) elicits a MT unfolded protein response in which the stress-activated transcription factor ATFS-1 is translocated to the nucleus, where it stimulates gene expression of MT chaperones. At low levels of stress, chaperones can reestablish cellular homeostasis by alleviating stress-induced damage and preventing the induction of apoptosis (33, 34). Persistent exposure to high levels of ROS increases the translocation of misfolded proteins to the mitochondria, which may cause ultrastructural damage. Together with MT uncoupling, this can impact cellular MT bioenergetic capacity and metabolic functions (31). Higher levels of stress may activate mitophagy, a process by which damaged mitochondria can be eliminated and replaced. For that, MT uncoupling can activate the PINK1-Parkin pathway, causing ubiquitination of MT membrane–associated proteins in damaged mitochondria. This ultimately stimulates lysosomal degradation and autophagy of mitochondria (35). In the most severe cases of lipotoxicity with high ROS production, the MT permeability transition pore is induced, leading to the release of cytochrome c into the cytoplasm. Consequently, this activates the caspase cascade, resulting in apoptosis (36) (**Figure 1**).

Importantly, when the capacity to metabolize or store FFAs is exceeded, this leads to increased production of intermediates like diacylglycerol and ceramides. Diacylglycerol activates serine/threonine kinases, which phosphorylate insulin receptor substrate 1, impairing insulin signaling. Also, ceramides inhibit phosphorylation of protein kinase B, inducing reduced trafficking of GLUT4 to the plasma membrane. Overall, this reduces insulin sensitivity in the affected tissues, which in turn exacerbates the lipotoxic state by inhibiting the suppression of lipolysis (37).

2.4. Disorders Associated with Lipotoxicity

Lipotoxicity plays a key role in different disorders in humans. A well-known example is nonalcoholic fatty liver disease. Here, the FFAs that are taken up in hepatocytes ultimately cause liver dysfunction with insulin resistance in the hepatocyte (38). Lipotoxicity is also known to damage pancreatic beta cells, contributing to the development and progression of type 2 diabetes (39). Increased lipid accumulation in cardiomyocytes also leads to metabolic cardiomyopathy, resulting in contractile defects of the heart and in severe cases even heart failure (40). Lipotoxicity in skeletal muscle disturbs insulin signaling; reduces insulin sensitivity, inducing insulin resistance; and reduces glucose uptake (41). In addition, both acute and chronic kidney diseases are aggravated in a lipotoxic state when excess FFAs bound to albumin are filtered. These FFAs are taken up by renal tubular cells and induce cellular dysfunction (42). Furthermore, in cases of neurological disorders and traumatic central nervous system injuries, lipotoxicity can exacerbate the cellular damage (43). Lastly, but most importantly for this review, lipotoxicity negatively affects the human and bovine ovarian follicle as well, resulting in altered endocrine signaling and follicular growth patterns and reduced oocyte quality (44–47).

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3. LIPOTOXICITY: OVARIAN FUNCTION, FOLLICULAR PHYSIOLOGY, AND THE CUMULUS–OOCYTE COMPLEX

3.1. Free Fatty Acid Uptake by the Oocyte

Numerous studies have shown that ovarian follicular fluid (FF) is an ultrafiltrate of serum, and therefore serum lipid concentrations are reflected in the FF. However, importantly, the concentration and type of lipids differ between serum and FF, as discussed below. This difference is mediated by the blood follicle barrier (BFB). The BFB consists of the vascular endothelium, subendothelial basal membrane, thecal interstitium, follicular basal membrane, and granulosa cells, and the passage of metabolites is determined by their charge and size (≤300 kDa) (48). Consequently, FFAs and high-density lipoproteins can pass the BFB. In contrast, chylomicrons, LDLs, and VLDLs cannot pass the BFB (49). However, LDLs and VLDLs are present in FF because they are secreted by granulosa cells (50). The FFAs in the FF are taken up by CD36 on the cumulus cells surrounding the oocyte [cumulus–oocyte complex (COC)] (51). The cumulus cells can convert saturated FFAs into monounsaturated FFAs through stearoyl-CoA desaturase (52) and also store these in lipid droplets to protect the oocyte (53). Nonetheless, the conversion and storage capacity are limited, and excess FFAs are passed to the oocyte through fatty acid binding proteins in the gap junctions between the cumulus cells and the oocyte. The oocyte itself can esterify FFAs into TGs and store them in lipid droplets as a protection mechanism (54). Increased bovine oocyte lipid content is associated with increased developmental competence after fertilization (55). Lipid droplets are closely linked to the oocyte mitochondria (6–10-nm distance in porcine oocytes), making them a metabolic unit capable of rapidly metabolizing lipids when necessary (56). Important interspecies differences exist regarding the oocyte's dependence on lipid metabolism. For example, pig oocytes contain approximately three times more lipids compared to cow and sheep oocytes (57). Human and mouse oocytes contain lower amounts of lipids compared to domestic species (58). Also, inhibition of β-oxidation does not negatively affect bovine and murine oocyte maturation due to their capacity to switch to carbohydrate metabolism. In contrast, partial inhibition of β-oxidation in pigs inhibits oocyte maturation (59). In summary, the BFB, cumulus cells, and oocyte lipid droplet formation form a strong first-line defense against lipotoxicity-induced damage (**Figure 2**). However, excess FFAs create a nutritive overload that may shift the metabolic preference of the oocyte to use more FFAs for energy production. This makes the oocyte more vulnerable to oxidative stress and lipid peroxidation, which leads to lipotoxicity at the oocyte level (21).

3.2. The FFA Composition of the Follicular Microenvironment and Its Impact on Oocyte Quality

Analyses of bovine and human FF have elucidated the role of FFAs in preovulatory follicles and how this can be altered by a lipotoxic state. Total FFA concentrations in FF ranged from 0.1 to 0.6 mM in cows (44), whereas in women the average total FFA concentration in the FF was approximately 0.2 mM (46, 60). In general, the most abundant fatty acids within the FFA fraction are the saturated fatty acids palmitic acid (PA) (23% in cows and 23–27% in humans) and stearic acid (SA) (13% in cows and 8–12% in humans) and the monounsaturated fatty acids oleic acid (OA) (33% in cows and 30–31% in humans) and linoleic acid (LA) (9% in cows and 16–25% in humans) (44, 46, 60). Importantly, FFA concentration differs between serum and FF of preovulatory follicles. For example, in the early pp period in cows, SA was significantly lower in FF compared to serum, whereas LA and α-linolenic acid (ALA) were higher in FF than in serum. At 44 days pp, PA, OA, and LA were increased significantly in the FF compared to serum, whereas SA was still significantly lower in FF (44). Also, FFA concentrations in the serum and FF are influenced

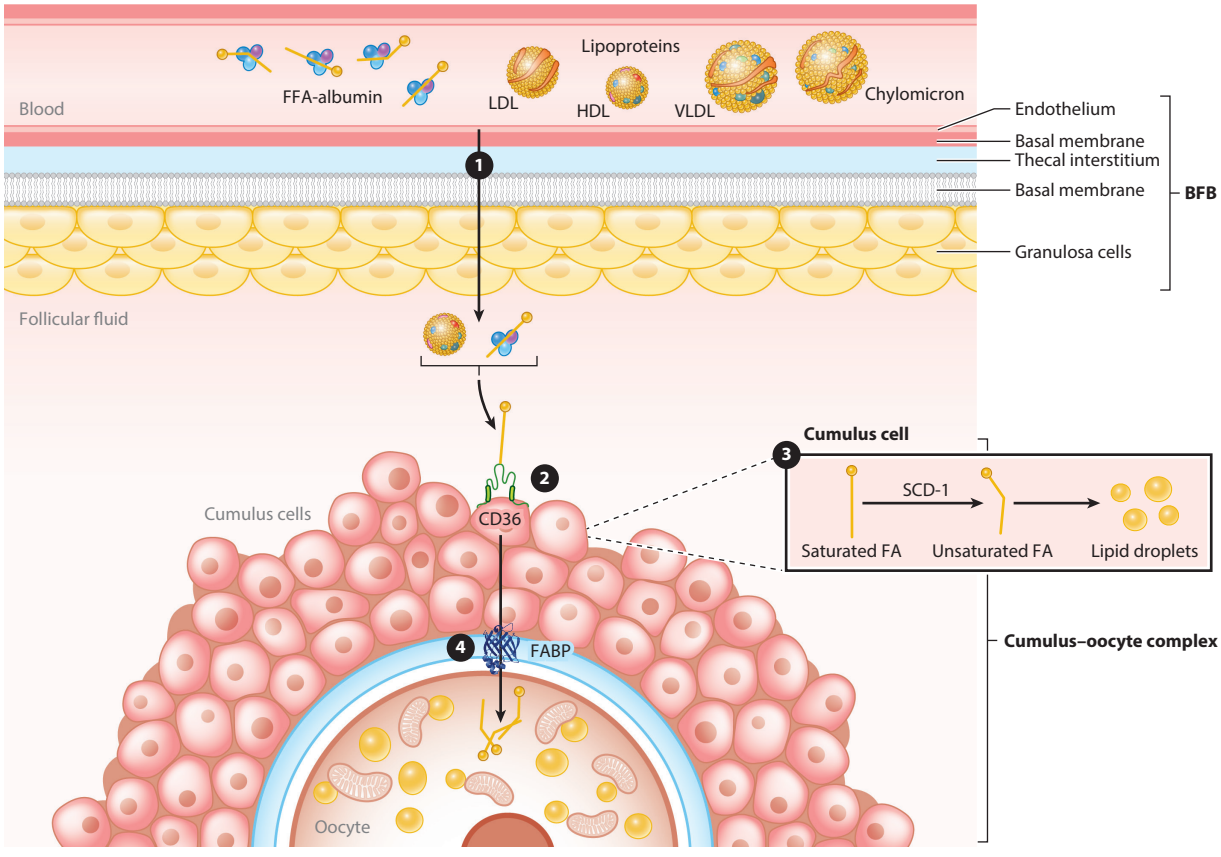

Figure 2

Schematic overview of mechanisms involved in the uptake of lipids from serum to follicular fluid and, ultimately, the cumulus–oocyte complex. Abbreviations: BFB, blood–follicle barrier; CD36, cluster of differentiation 36; FA, fatty acid; FABP, fatty acid binding protein; FFA, free fatty acid; HDL, high-density lipoprotein; LDL, low-density lipoprotein; SCD-1, stearyl-CoA desaturase 1; VLDL, very low-density lipoprotein.

by dietary fat intake. For example, when rumen-protected polyunsaturated fatty acids (e.g., ALA) were supplemented in the diet of cows, this was also increased in the serum and FF (61). Analysis of human follicles demonstrated the role of the maternal metabolic state in the FFA composition of the FF. In this study, increasing BMI in women was associated with higher FFA concentrations in the FF (60).

Regarding the impact of FFAs on oocyte quality, many studies have shown that elevated levels of saturated fatty acids are detrimental, whereas mono- and polyunsaturated fatty acids are beneficial. More specifically, higher concentrations of PA and SA in human FF were associated with failure of embryo cleavage after fertilization (62). In another human study, PA was found to be the only FFA in FF that was negatively correlated with pregnancy success in women undergoing intracytoplasmic sperm injection, and SA in FF was negatively correlated with oocyte maturation rate (63). On the other hand, OA concentrations in FF were positively correlated with pregnancy success (63), and LA in human FF was associated with improved fertility rates (64). Oocytes collected from obese women exhibit abnormal morphology, lower maturation rates, lower embryo yields following in vitro fertilization (IVF), higher rates of embryo fragmentation and aneuploidy, and lower

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pregnancy rates (45, 46, 63, 65). In in vivo mouse studies, oocytes collected from diet-induced obese mice exhibited high intracellular lipid accumulation, oxidative stress, and subsequently lower embryo viability and quality (66–70). Importantly, although the studies in obese women and diet-induced obese mice most closely resemble the real-life situation, parameters beyond lipotoxicity affect oocyte quality. For example, in such individuals, the hypothalamic–pituitary–ovarian axis is affected, leading to endocrine changes that alter processes like folliculogenesis and ovulation (71). The primary importance of reduced oocyte quality in the pathogenesis of subfertility associated with obesity was illustrated in a study where pregnancy rates were restored in obese women when donor oocytes from healthy mothers were used (3). Furthermore, other studies in mouse models found that detrimental effects persisted in embryos from obese mothers even when these embryos were transferred to the uterus of control-fed mice (72, 73).

3.3. Mimicking the Lipotoxic Follicular Microenvironment In Vitro

Several studies have specifically mimicked the increased FFA concentrations in the FF associated with lipotoxicity during oocyte in vitro maturation (IVM). The concentrations are often based on in vivo studies. For example, in a study in cows, the highest FFA concentrations during NEB were 133 μM PA, 67 μM SA, and 200 μM OA, compared to basal concentrations of 44.3 μM PA, 22.3 μM SA, and 66.7 μM OA (44). In studies in our lab, COCs were matured in a combination of lipotoxic FFAs (150 μM PA, 75 μM SA, and 200 μM OA) and then fertilized and cultured in control conditions until the blastocyst stage (day 7). At the end of IVM, oocytes displayed increased mRNA expression of genes related to REDOX regulation and reduced blastocyst rates after IVF (74). In the resulting blastocysts, besides increased apoptosis rates, the metabolism was altered through reduced oxygen, pyruvate, and glucose consumption; upregulated lactate consumption; and increased amino acid metabolism (75). Other studies exposed oocytes to 150 μM PA only during IVM. This induced proteomic changes related to unfolded protein response, MT dysfunction, lipid metabolism, and oxidative stress in oocytes. Also, ROS concentrations were increased in the oocytes (76). In another study, blastocysts originating from PA-exposed oocytes were transferred to healthy cows, and embryos were recovered again seven days later. In the recovered day-14 embryos originating from PA-exposed oocytes, growth retardation was still observed, together with altered cellular metabolism, and transcriptomic changes in pathways related to redox regulation, apoptosis, cell growth, interaction and differentiation, energy metabolism, and epigenetic regulatory mechanisms (77). Also, exposure of porcine oocytes to 500 μM PA during IVM reduced oocyte maturation rate, elevated ROS levels and reduced ATP levels (78), increased lipid content, and altered histone modification patterns in the oocytes (79). Furthermore, 67 μM SA during bovine IVM significantly reduced oocyte maturation rate, embryo cleavage rate, and blastocyst rate and increased the proportion of apoptotic cells in the COC (44). In another study, 75 μM SA during bovine IVM increased mRNA expression of metabolism- and oxidative stress–related genes (74). In contrast to PA and SA, OA has beneficial effects on the COC. Use of 500 μM OA increased bovine oocyte neutral lipid accumulation during IVM, and 250 μM OA could counteract the negative effects of both 250 μM PA and SA on lipid accumulation and developmental competence (80). In contrast to observational human studies (64), exposure of bovine oocytes to LA during IVM reduced oocyte maturation rate and embryo developmental competence with increasing concentrations (50 μM, 100 μM, and 200 μM) (81). Although ALA is present at lower concentrations in FF (44, 60), it also has important protective effects against FFA-induced lipotoxicity. For example, adding 50 μM ALA to the bovine IVM medium alleviated the negative effects of combined FFAs (150 μM PA, 75 μM SA, and 200 μM OA) on cumulus cell viability and embryo development and quality (82). Taken together, not only the concentration but also the type and degree of saturation and ratios of FFAs determine the impact on oocyte and embryo quality.

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An important limitation of this IVM model is that the fatty acids require a solvent, such as ethanol or dimethylsulfoxide (DMSO), to be dissolved in the aqueous maturation medium, in contrast to the in vivo situation. Low concentrations of DMSO (0.5%) improved blastocyst rate and quality, whereas a high DMSO concentration (1% or 10%) resulted in no blastocyst development (83). Adding 0.1% ethanol to the bovine maturation medium resulted in increased oocyte MMP and H3K9me3 levels at the 8-cell stage (84). Therefore, FFA-exposed oocytes or embryos must be compared to a control group exposed to an equal concentration of solvent. Another important limitation is that IVM comprises only the final +/−24 h (species dependent) of oocyte maturation before fertilization. This is in stark contrast to the in vivo situation, where the COC is exposed to the lipotoxic follicular environment starting from the tertiary follicle phase, in which a fluid-filled antrum is formed (85). Therefore, some studies have also investigated the role of long-term FFA exposure during folliculogenesis on oocyte and embryo quality. For example, in vitro exposure of murine ovarian follicles to a pathophysiologically relevant mixture of FFAs (112 mM SA, 210 mM OA, and 230 mM PA) for 12 days reduced oocyte developmental competence (86).

3.4. Which Stages of Folliculogenesis Are Vulnerable to Lipotoxicity?

Although most studies attributed the lipotoxicity in oocytes to alterations in the lipid composition of the FF in the preovulatory follicles, whether and how the microenvironment of pre-antral and early-antral follicles is affected by a lipotoxic state are currently not known. However, some studies suggest that oocytes might already be sensitive to lipotoxicity before antrum formation. The whole period of folliculogenesis takes approximately three weeks in mice and three to four months in humans and cattle (87, 88). Dairy cows can suffer from NEB during the first three weeks pp, and the first artificial insemination takes place 50–60 days pp, when serum FFAs are usually normal again. However, surprisingly, cows that suffered from severe NEB still had lower pregnancy rates, suggesting that an oocyte, at the moment of ovulation, may have been affected during early folliculogenesis (9). Also, a more recent study in dairy cows showed that high FFA concentrations associated with maternal metabolic stress in the pp period can have long-term effects on granulosa cell functions in preovulatory follicles at the time of breeding (47). Studies using a mouse model demonstrated that diet normalization and caloric restriction for six weeks after HFD did not completely alleviate the MT ultrastructural abnormalities (89), and the increased oocyte lipid droplet volume, elevated MMP, reduced pericortical distribution of active mitochondria, and negative effect on oocyte developmental competence after IVF were all persistent after four weeks of diet normalization (67). All these studies provide strong suggestions for an affected primordial follicle pool due to lipotoxicity. However, this has not yet been specifically investigated.

4. MITOCHONDRIA AS SENSORS

4.1. Oocytes Heavily Rely on Mitochondria

Mitochondria are essential organelles for oocyte function (90). Oocyte mitochondria are relatively small (0.5–1 μm) and have an immature, spherical morphology with only a few cristae (91), making them unable to produce high amounts of ATP (92). But, because oocytes and cleavage-stage embryos lack the enzymatic machinery for glycolysis, they are completely reliant on the mitochondria for ATP production (93). Therefore, from the primordial follicle stage to the mature oocyte stage, mitochondrial DNA (mtDNA) copy number increases from approximately 200 to more than 200,000 (94, 95). This makes the mitochondria the most abundant cytoplasmic organelle (96). An optimal supply of ATP for the mature oocyte is required for several important developmental processes such as kinase activation, cytoskeletal dynamics, and cortical granule activation (95, 97, 98). Furthermore, mitochondria also control cellular metabolism, redox regulation, and

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Ca^{2+} homeostasis (99). As the primary site of cellular respiration, mitochondria are the main cellular target for environmental stressors, such as lipotoxicity (100). Although mature oocytes contain a high number of mitochondria and can cope with stressors through several adaptive mechanisms (31, 33), mitochondria can still become dysfunctional when a certain threshold is reached.

4.2. Mitochondria: The Bridge Between Lipotoxicity and Reduced Oocyte Quality

Human, murine, and bovine studies have shown that lipotoxicity-induced MT dysfunction in oocytes is characterized by altered MMP (70), increased MT biogenesis (also called mitogenesis) (68), changes in mtDNA copy number (72, 95), deficient β-oxidation (101), altered ATP production (102), altered MT distribution (103), and MT ultrastructural abnormalities (66, 72). Also, in the bovine in vitro model, after exposure to 150 μM PA, 30% of the lipotoxicity-induced proteomic changes in oocytes were linked to MT functions (76). Consequently, due to the mitochondria's importance and high abundance, several very specific oocyte defects occur that lead to reduced developmental competence. This includes abnormal spindle morphology and chromosomal segregation, aneuploidy, failure of fertilization, increased embryo cell fragmentation, and developmental arrest (66, 76, 104, 105). We observed in the bovine in vitro model that embryos produced from oocytes exposed to 150 μM PA exhibit persistent reduced MMP in 4-cell embryos and increased mRNA expression of markers related to MT unfolded protein response (HSP10, HSP60) and MT biogenesis (TFAM) in blastocysts (104).

In somatic cells, dysfunctional mitochondria can be cleared specifically by mitophagy or rejuvenated by mitogenesis. Remarkably, these processes are not yet understood fully in oocytes. In a study using CCCP (carbonyl cyanide *m*-chlorophenyl hydrazone, 10 μM for 2 h) to induce MT depolarization and initiate mitophagy in murine oocytes, mtDNA copy number remained stable, and no co-localization of mitochondria with autophagosomes was observed despite increased PINK1 expression. This suggests that the dysfunctional MT population was not eliminated after introducing a specific MT insult (68).

5. CARRYOVER EFFECTS ON OFFSPRING HEALTH

5.1. Lipotoxicity During Different Developmental Periods

Besides the effect of lipotoxicity on maternal fertility through reduced oocyte quality, offspring can also be affected when the embryo survives. The DOHaD (developmental origins of health and disease) theory postulates that stressors, like a disturbed metabolic environment, during different sensitive windows of the preconception and pre- and perinatal period are responsible for the development of noncommunicable diseases in later life (106). Firstly, the preconception period is critical. We already discussed how oocytes are exposed to maternal metabolic stress during folliculogenesis. In addition, not only the oocyte but also the preimplantation embryo can be subjected to a lipotoxic insult. After fertilization, the embryo enters the oviduct, in which the oviductal fluid (OF) reflects the maternal metabolic state and forms the microenvironment of the preimplantation embryo (107). A study investigating bovine OF showed that OF and serum FFA concentrations did not differ significantly and were positively correlated (108). When serum from cows fed a diet rich in saturated fats was added to the in vitro culture (IVC) medium, blastocyst rates decreased, and mRNA expression of genes related to apoptosis and oxidative stress was altered, compared to when serum from cows fed a diet rich in unsaturated fatty acids was added (109). In the same study, unsaturated fatty acid–rich serum did not affect embryo development and quality (109). Also, research in our lab has shown that murine oviductal epithelial cells already exhibit increased expression of genes related to oxidative stress and ER stress after three days of

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HFD, and this persisted until 16 weeks of HFD (110). Analysis of the murine oviductal epithelial cell lipidome after a HFD revealed that differentially regulated lipids were identified already after three days of HFD feeding, and the number of differentially regulated lipids gradually increased and became more persistent with longer HFD exposure (111). IVC of murine embryos with PA (25–100 μM) resulted in concentration-dependent decreased embryo development, which could be counteracted with increasing OA concentration (50–250 μM) (112). After the preimplantation period, the embryo can also be affected by maternal metabolic stress via the uterine microenvironment (113, 114), and these effects also extend after birth during lactation due to changes in milk composition (115). Although a maternal lipotoxic environment can affect the offspring during all aforementioned periods, the involved mechanisms differ, as discussed below.

Human studies have generated many interesting insights (116). For example, the offspring of obese women were shown to have an increased risk of several diseases, like obesity, type 2 diabetes, asthma (117), cardiovascular diseases (118), and neurodevelopmental disorders (119). However, these are all observational studies and therefore do not contribute to understanding the importance of each of the developmental periods and the responsible mechanisms of the transgenerational effects. On the other hand, specifically designed animal studies have provided more specific insights into dissecting the importance of each exposure window. For example, in the mouse, a maternal HFD before and during pregnancy and during lactation induces offspring adipocyte hypertrophy, artery endothelial dysfunction, hypertension, and elevated fasting insulin and glucose (120). In another study, maternal HFD during pregnancy and lactation caused increased body mass, adipocyte hypertrophy, and insulin resistance in offspring (121). Interestingly, another study also used a model in which mothers were fed a HFD during pregnancy and lactation, but this was compared to a group where mothers received a HFD only in the preimplantation period (from conception until embryonic day 3.5). A HFD during pregnancy and lactation altered offspring energy metabolism and brain physiology. But, strikingly, only three days of HFD exposure during the preimplantation period was sufficient to generate these effects, albeit less severely (122). This highlights the critical importance of the earliest stages of embryo development in shaping offspring health. Importantly, offspring oocyte quality can be hampered as well. In studies where mother mice were fed a HFD from the preconception period until weaning, offspring oocytes displayed increased MT ultrastructural abnormalities (123), altered MT localization, increased MMP, decreased MT mass, ATP content, and citrate content and changes in global DNA methylation levels (124). In contrast to the aforementioned studies, which used inbred C57BL/6 mice as models, we have validated the use of an outbred Swiss mouse model to investigate this. Commonly used inbred strains, like C57BL/6, are characterized by reduced fertility (125) and increased oocyte cellular stress and MT ultrastructural abnormalities compared to Swiss mice (66). On the other hand, outbred strains like Swiss are more fertile and have a higher genetic variability, and their metabolism and oocytes seem more responsive to HFD. Therefore, they are more suitable for research involving oocyte and embryo quality and more closely resemble human physiology (66). Strikingly, research in our lab using the outbred Swiss model showed that oocytes of offspring born to obese mothers exhibit limited changes in MT ultrastructure without altering MMP, active MT distribution, MT DNA copy numbers, or ROS production. Corresponding changes in the expression of MT proteins of complex III and V suggest that changes induced by maternal obesogenic diet appear to be adaptive and compensatory rather than dysfunctional (126).

5.2. Potential Mechanisms for Propagation to the Offspring

The first possible mechanism to explain offspring effects, if they exist, is oocyte MT dysfunction. Defective oocyte mitochondria are suggested to be transferred to the embryo, as they are exclusively maternally inherited, mitophagy in oocytes seems to be inactive, and MT replication

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is initiated only from the blastocyst stage onward (68, 127). In the blastocyst, the onset of MT replication is specific to the trophectoderm cells, which is crucial for implantation. In the inner cell mass, MT replication starts only when these cells have received signals to differentiate into specific cell types (128). Therefore, accumulating dysfunctional oocyte mitochondria could persist in the embryo. This ultimately affects the metabolism, cell differentiation, and survival of the embryo, even if it is transferred to a healthy uterus (77). Importantly, many studies suggest the existence of the germline MT bottleneck, which avoids transmission of defective mitochondria through the germline. During the peri-implantation period, only germline cells with a healthy MT population survive, and hence transmission of defective mitochondria to the offspring oocytes is minimized (129, 130). A recent study states that this bottleneck theory is unlikely and proposes a new theory based on computational modeling to explain how transmission of MT dysfunction through the germline is avoided (131). During fetal development, primordial germ cells become organized in germline cysts containing five to eight oogonia that are connected through cytoplasmic bridges. Then, the so-called nurse cells transfer their mitochondria through the cytoplasmic bridges to the Balbiani body of one given oogonium that becomes the primary oocyte. As a result, dysfunctional mitochondria remain in the nurse cells, and the primary oocyte contains the healthy mitochondria. In the end, the nurse cells undergo apoptosis (131). However, this modeled hypothesis requires validation with empirical data. Consequently, as the transfer of aberrant mitochondria during the periconceptional period is subject to discussion, what other mechanisms can explain the downstream effects on offspring health?

Another explanation for the observed effects on offspring health is that mitochondria affect epigenetic reprogramming during folliculogenesis and early embryo development. During these periods, the oocyte and embryo undergo an extensive reprogramming of epigenetic marks, like DNA methylation and histone modifications, and genomic imprinting (132). This is crucial to acquire totipotent cells as an essential requirement for proper development. Also, this ensures that epigenetic marks that accumulate in the parents are erased (133). The dynamic nature of the reprogramming makes the oocyte and embryo epigenome sensitive to changes in the microenvironment. When epigenetic marks are established incorrectly or even erased in oocytes and embryos, this can be irreversible, causing permanently altered expression of developmentally important (imprinted) genes, developmental aberrations, and compromised postnatal health (134). Several studies have shown that diet-induced obesity in mice significantly modifies global DNA methylation and histone modifications in fully grown oocytes (135, 136). In somatic cells, MT intermediate metabolites, like ATP, *S*-adenosyl methionine, and acetyl-CoA, can serve as substrates and/or cofactors for epigenetic modifications and epigenetic regulatory enzymes (like DNA methyltransferase and ten-eleven translocation enzymes) (137). This interplay has been referred to as metaboloepigenetics (138). However, this link is not evident in oocytes and early embryos due to the specific morphology and function of the mitochondria and the dynamic nature of the epigenetic landscape. Only recently, some studies have started to further elucidate this link. Research conducted in our lab in a bovine in vitro model demonstrated that oocyte IVM and/or embryo IVC under lipotoxic conditions resulted in MT dysfunction in early embryos associated with changes in DNA methylation patterns and histone modifications (77, 139). Furthermore, by specifically inhibiting oocyte MT ATP production using 5 nmol/l oligomycin A only temporarily during bovine IVM, DNA methylation and histone modification patterns were altered in oocytes and early embryos, despite restoration of MT functions in the embryos. With this research, a direct link between oocyte ATP production and oocyte and embryo epigenetic programming has been suggested for the first time (140). Elsewhere, it was found that oocyte mtDNA mutation affected embryo DNA methylation levels, and that this could be restored by supplementation of α-ketoglutarate during murine embryo culture (141). Also, the deletion of MT

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fission factor Drp1 in oocytes altered oocyte DNA methylation and histone methylation patterns (142). Although knowledge on this topic is growing, more research is required to fully understand mechanisms linking mitochondria with epigenetic programming pathways in oocytes and preimplantation embryos.

During the intrauterine and lactation period, maternal metabolic and endocrine alterations can directly affect the metabolic and endocrine regulation of the respective fetus and offspring. Consequently, this can generate de novo epigenetic changes in offspring. Ultimately, this can program offspring in such a way that metabolic, endocrine, and epigenetic alterations persist after weaning even when offspring are no longer directly exposed to a lipotoxic environment. This concept is referred to as developmental programming (143). In case oocytes of offspring (F1) are still affected by adulthood, this can induce carryover effects to the next generation (F2). These are called intergenerational effects, as F1 (in utero) and F2 (germ cells of F1 in utero) are directly exposed to the insult. Transgenerational effects occur when the F3 generation, which has not been directly exposed to the insult, is still affected by the exposure of the F0 generation to the lipotoxic insult (123, 144). Importantly, developmental programming should ensure that the fetus is optimally adapted to its postweaning environment. However, if the offspring's diet after weaning does not match the maternal diet, the adaptive responses can be maladaptive (145). This is also what was observed in a recent study in our lab. Here, outbred mice were fed a HFD for seven weeks before mating, during pregnancy, and during lactation (146). Afterward, offspring were maintained on a HFD or switched to a control diet. A maternal HFD was associated with increased expression of MT complex markers and MT ultrastructural abnormalities in offspring muscle but had only limited effects on the offspring metabolic profile. However, in pups born to HFD-fed mothers, no increased sensitivity to an offspring HFD was observed. Even more interestingly, their metabolic profile was healthier compared to offspring fed a HFD coming from lean mothers. This suggests that, in this outbred model, a maternal HFD induces an adaptation of the offspring in such a way that they are more capable of coping with a HFD themselves (146).

6. TREATMENT FOR LIPOTOXIC EFFECTS IN OVARIAN CELLS AND OOCYTES

As explained earlier, we currently have a good understanding of the pathogenesis of lipotoxicity and how this affects oocyte quality. In recent years, an increasing amount of research has emerged focusing on the treatment of these oocytes. The proposed strategies can be divided into two groups: preconception treatments and curative treatments.

6.1. Preconception Treatments

Preconception treatments focus on "winning the battle before the war has begun" (147, p. 612) by supporting the oocyte during the preconceptional period. In vitro, the final maturation period of the oocyte can be used as an opportunity to rescue metabolically compromised oocytes. A possible approach is the use of an ER stress inhibitor, like salubrinal, which has been proven to prevent oocyte MT dysfunction and reduce embryo development after FFA exposure (148). Also, supplementation of L-carnitine during bovine IVM was demonstrated to protect the COCs against FFA-induced lipotoxicity (149). Another option can be to reduce oxidative stress and concomitant MT dysfunction by supplying antioxidants (AOs). Several studies established that supplying AOs during IVM is beneficial for embryo quality (150, 151). However, their application is limited because their capacity to reduce oocyte oxidative stress or restore MT functions during ongoing metabolic stress is insufficient (152, 153), and only water-soluble AOs can be used without the need for solvents (11). Therefore, mitochondria-targeted AOs might be more suitable. In many

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tissues and cell types, these have been shown to improve disorders related to metabolic syndrome (154). A study in our lab evaluated the effect of MitoQ, which consists of coenzyme Q10 (CoQ10) bound to a cationic carrier to ensure translocation to the mitochondria, during bovine IVM under lipotoxic conditions. Lipotoxic effects on oocyte and blastocyst development and MT function were alleviated by adding MitoQ during IVM (155). Another drug that accumulates in mitochondria, and has proven its efficiency, is BGP-15. This improves MT function and mtDNA content in oocytes from obese mice (69). Although the in vivo use of such oocyte MT-targeted therapies is urgently needed and has a high potential, their implementation is hampered because IVM is not used routinely in the human IVF clinic.

Metabolically compromised women are often advised to lose weight or exercise before conception to increase pregnancy odds (156, 157). Nonetheless, the results of clinical trials are conflicting. Whereas several studies fail to show improved fertility after preconception weight loss (158, 159), others have proven that preconception weight loss in overweight or obese women undergoing assisted reproductive technologies resulted in a reduction of required cycles, an increased number of embryos available for transfer, and improved pregnancy rates (160). However, even there, results regarding which type of intervention is preferred are conflicting. How much weight loss can be considered beneficial, and for how long should this intervention last? These studies are also often confounded by a lack of power, high drop-out rates, and not taking the patient's history and lifestyle into account. Consequently, univocal scientific guidelines are lacking (161). This emphasizes the need for well-designed in vivo animal studies, as these are much less restricted by practical and ethical limitations. In dairy cattle, strategies to avoid excessive weight loss and NEB during the transition period succeeded in protecting oocyte quality (162). However, in a study using an inbred mouse model, the negative effects of a HFD on oocyte quality were not reversed by diet normalization for eight weeks (163). We demonstrated that preconception diet normalization or caloric restriction for four to six weeks in obese outbred mice could alleviate the negative effects on oocyte quality depending on the type and duration of preconceptional care intervention. This included (partially) improved oocyte MT ultrastructure, MMP, and ROS levels (67, 89). In a follow-up study to assess the health of the offspring of these mice, we showed that maternal HFD increased offspring serum insulin levels and had a limited effect on the percentage of oocyte MT ultrastructural abnormalities and MT size in the offspring. Also, global DNA methylation levels were increased, but proteins related to cellular stress were downregulated in the offspring oocytes. Preconceptional diet normalization in the mother for four weeks could prevent most of the alterations in the offspring, although this was the case for only a few parameters in the group that received preconception caloric restriction (164). This further highlights the importance of protecting oocyte quality during the preconceptional period.

6.2. Remedial Treatments

Remedial strategies aim to restore oocyte and embryo quality after the lipotoxic damage has occurred. We have demonstrated that, following PA exposure during bovine IVM, MitoQ added to embryo culture media alleviates detrimental effects on embryo development and MT function (104). However, consequences for subsequent development and offspring health are not known and should therefore be evaluated in future research.

7. CONCLUSIONS

A great amount of evidence has established that lipotoxicity negatively affects ovarian function and the follicular microenvironment. For the oocyte, this is associated with reduced developmental competence, altered metabolism, MT dysfunction, and epigenetic alterations. Reduced oocyte

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quality is of primary importance in the pathogenesis of subfertility. Besides this effect on oocyte quality, offspring health might also be at risk when the oocyte is fertilized, but the described effects and mechanisms appear to depend on the research model used. Our fundamental understanding of the mechanisms by which lipotoxicity can reduce oocyte quality has also led to several promising targeted treatments. However, these are not yet tested or routinely used in practice. Therefore, much work remains to be done to fully understand how lipotoxicity in the oocyte can ultimately affect offspring health and how this might be prevented or treated.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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